Particle Physics
Winter Quarter 2026
Physics 130A

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Physics 317

Lecture: M,W,F 9:00-9:50 AM in Giedt 1007

Textbooks: LN: Lecture Notes (Course website)
G: Introduction to Elementary Particles (2nd), Griffiths.
T: Modern Particle Physics, Thomson. (Optional)
An Intro. to Tensors and Group Theory for Physicists, Jeevanjee. (Optional)
(Jeevanjee is available online at the library)

Course TA: TBD tbd@ucdavis.edu

Office Hours: Mulhearn: Monday 10:00-11:00 AM PHY 317

Homework: From 5-10 homework problem sets.
Midterm Exams: Two In-Class Midterms
Final Exam: March, 16 2026 3:30 PM

Course Description:
Properties and classification of elementary particles and their interactions. Experimental techniques. Conservation laws and symmetries. Strong, electromagnetic, and weak interactions. Introduction to Feynman calculus.

Contacting the Instructor:
I have posted my email, but that is mainly so that you can contact me after the course has ended. During the course, the most reliable way to reach me is by talking to me after lecture or coming to office hours. If you must reach me remotely, please use the course website instead of email.

Reading:
This course will be focused on Griffiths, and the lecture notes, which draw from Thomson and Jeevanjee. **You are expected to read Giffiths chapter one on your own.** We will cover only a subset in lecture, but the material in the text is accessible.

Homework:
There will be 5-10 homework assignments, which you will submit to gradescope. You may consult with other students but your submitted work must be your own. Your homework will be graded on (sincere) effort only. The grading scheme for the course assumes students will be near 100% on homework. One late homework may be submitted at (or before) the final exam for full credit. There will be 5-10 homework assignments.

Double-Starred Problems:
Some material is accessible enough for you to work out on your own, but also too important to miss out on. Therefore, some homework assignments will include a "double starred" problem (**). These are treated as ordinary homework problems at first (so graded on effort only). After the solutions are posted, you may resubmit a neat, complete (including explanations in your own words), and correct hand-written solution at or before the next assignment or exam. (Do not worry if English is a second language for you, you comments will be graded for content not for grammar.) **You must correctly complete all double-starred problems (even if late) in order to pass the course.**

Exams:
Exams will be in class, paper and pen or pencil only, with no equation sheet, no internet, and no calculators. The exam will start with a quick answer, multiple choice part. No practice exams will be provided. Working through the homework problems for yourself will be best preparation for the exams.

Grades:
Final grades will be a mixture of homework (10%) and exams (90%). The weighting may be adjusted to avoid the need for a curve. For this reason, it is assumed that your homework scores (graded on effort only) will be near 100%.

Course Schedule:
This is a preliminary schedule which will evolve as the course progresses.
You are expected to read and comprehend Griffith's Chapters 1 and 2 on your own, and that material will be included on the exams.

Week	Dates	Topics	Reading
1	Jan 5	Intro to Particle Physics	G: 2.1-2.5 T: 1.1
	Jan 7		
	Jan 9	Vectors and dual vectors	LN: R.1-2
2	Jan 12	Tensors and metric duals	LN: R.3-4
	Jan 14	Invariance of space-time interval	LN: R.5
	Jan 16	Four-vectors	LN: R.6 G: 3.1-3.2
3	Jan 21	The Lorentz group	LN: R.7-8
	Jan 23	Relativistic kinematics	LN: R.9 G: 3.3-3.5
4	Jan 26	Examples	
	Jan 28	Decay rates and scattering	G: 6.1
	Jan 30	The interaction picture	LN: FGR.1-3
5	Feb 2	Fermi's golden rule	LN: FGR.4-5 G: 6.2
	Feb 4	Lorentz covariance	LN: FGR.6-7
	Feb 6	Midterm (Tentative)	
6	Feb 9	Feynman rules for a toy model	G: 6.3
	Feb 11	The Klein-Gordan equation	LN: D.1-2
	Feb 13	The Dirac equation	LN: D.3-4
7	Feb 18	Anti-particles	LN: D.5-9
	Feb 20	Spin	LN: D.10-12
8	Feb 23	Bilinear Covariants	LN: D.13
	Feb 25	TBD	
	Feb 27	TBD	
9	Mar 2	TBD	
	Mar 4	TBD	
	Mar 6	TBD	
10	Mar 9	TBD	
	Mar 11	TBD	
	Mar 13	TBD	